Počítačové cvičení č. 5
Zvýrazňování 2
Lokální 2D operátory - vyhlazování

Číslicové zpracování obrazů

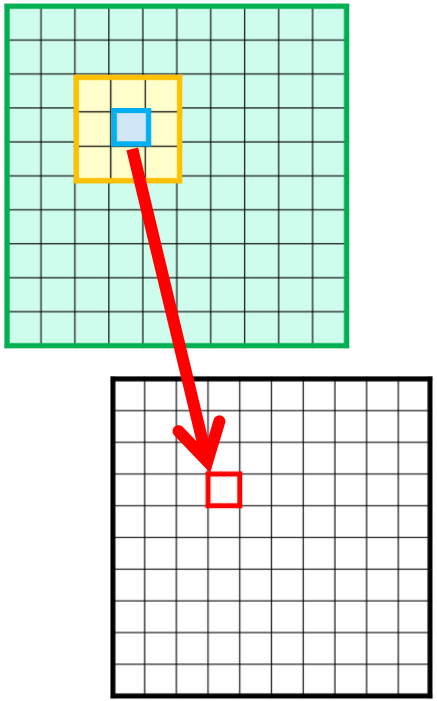
Petra Nemčeková, Kateřina Krejčí
Ústav biomedicínského inženýrství

Typy operátorů

- **Bodové operátory** – pixel ve výstupním obrazu je ovlivněn pouze pixelem na odpovídající pozici ve vstupním obraze

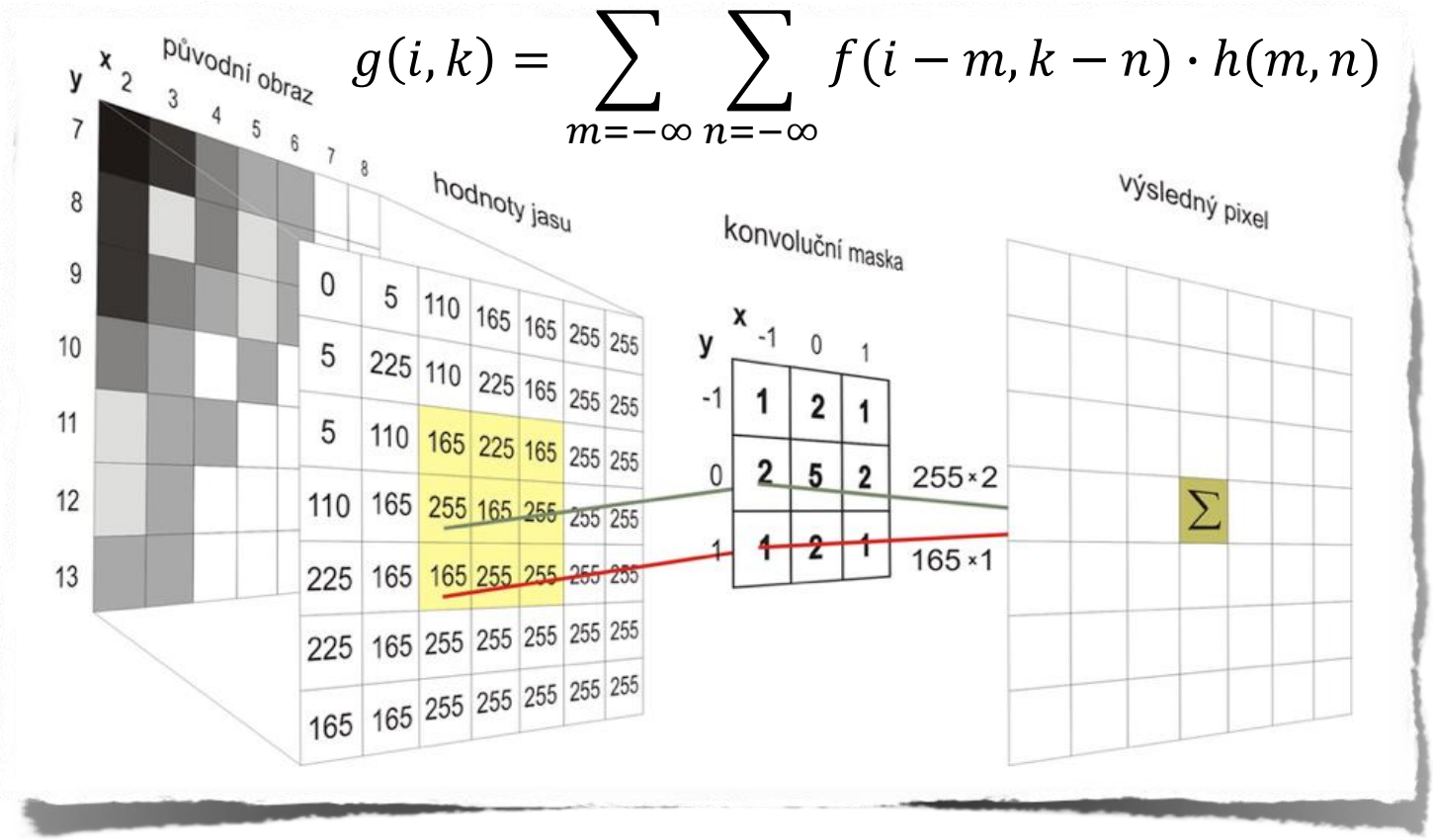
- **Lokální operátory** – pixel výstupního obrazu je ovlivněn pouze omezeným okolím pixelu ve vstupním obrazu (velikost okolí je dána velikostí masky).

- **Globální operátory** – každý pixel výstupního obrazu je ovlivněn všemi pixely vstupního obrazu.



2D Diskrétní konvoluce

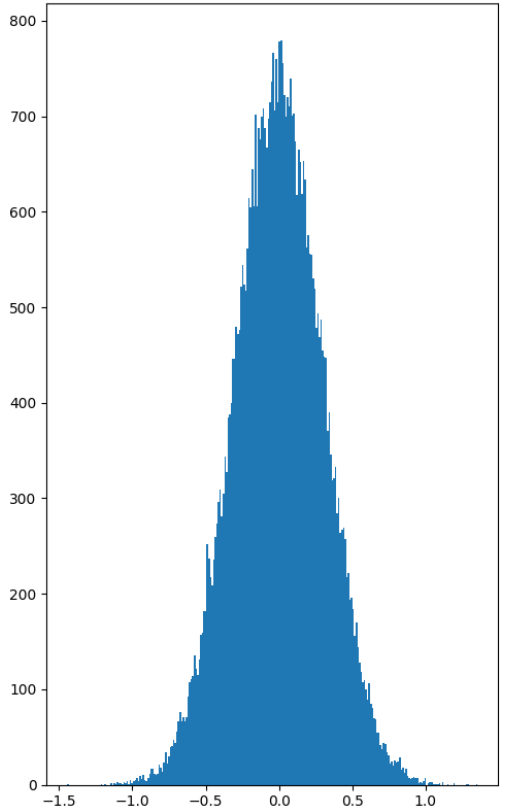
$$g(i,k) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} f(i-m, k-n) \cdot h(m,n)$$



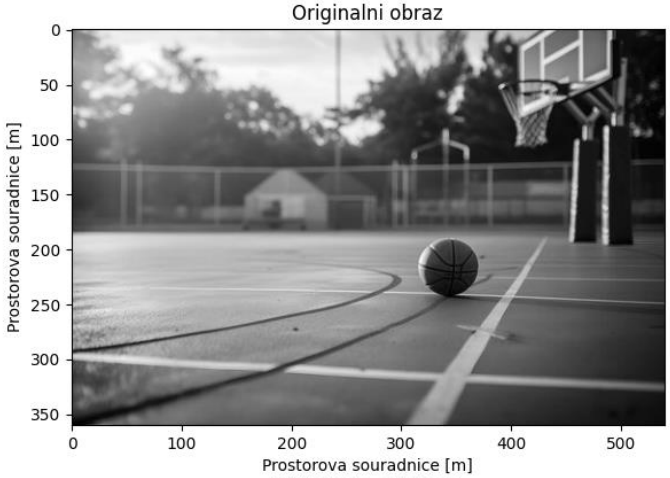
Gaussovský šum

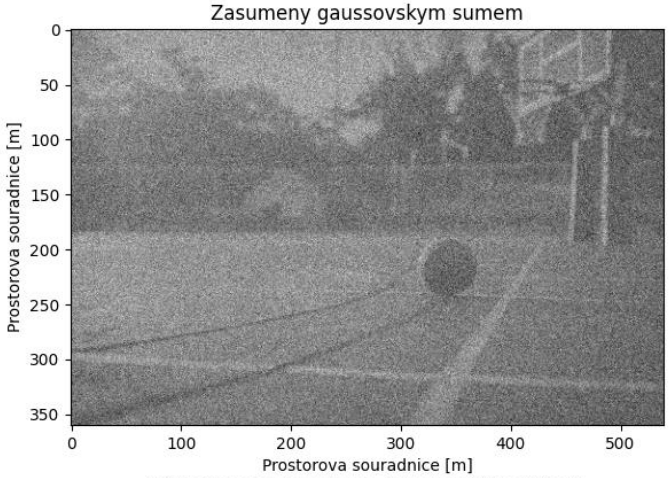


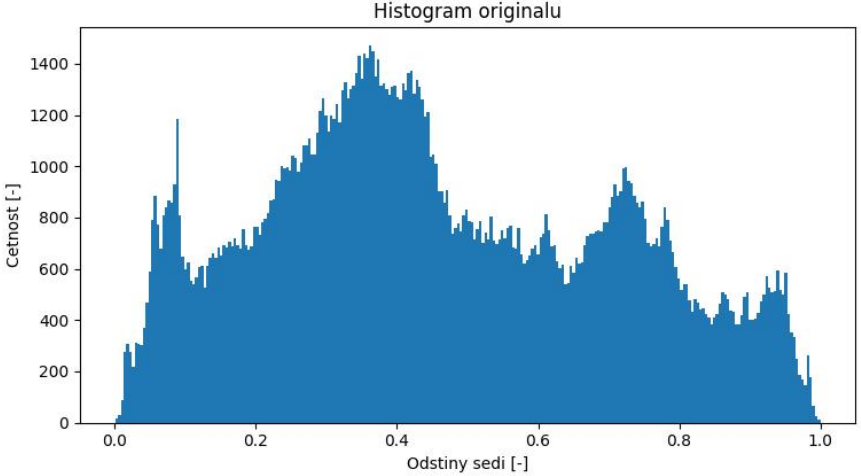


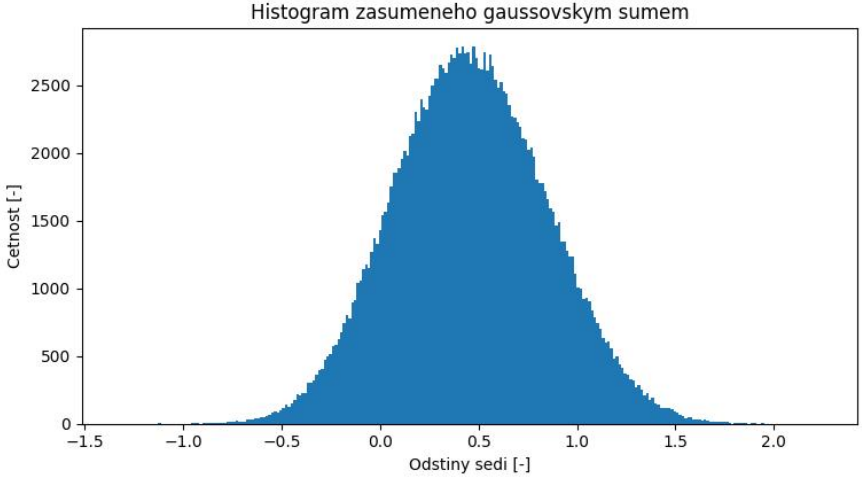


Úkol 1


Originalni obraz
Prostorova souradnice [m]
Prostorova souradnice [m]


Zasumeny gaussovskym sumem
Prostorova souradnice [m]
Prostorova souradnice [m]


Histogram originalu
Cetnost [-]
Odstiny sedi [-]

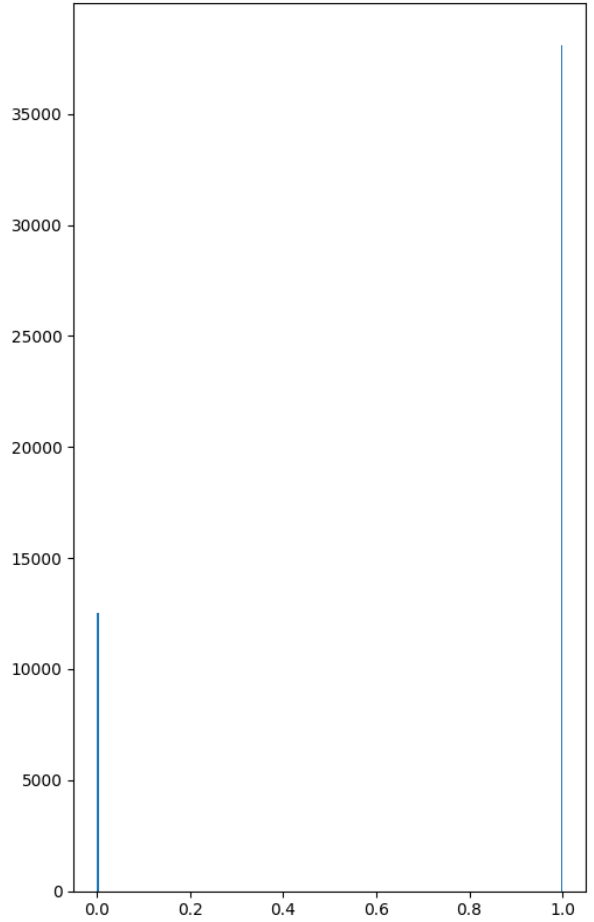

Histogram zasumeneho gaussovskym sumem
Cetnost [-]
Odstiny sedi [-]

Impulsní šum

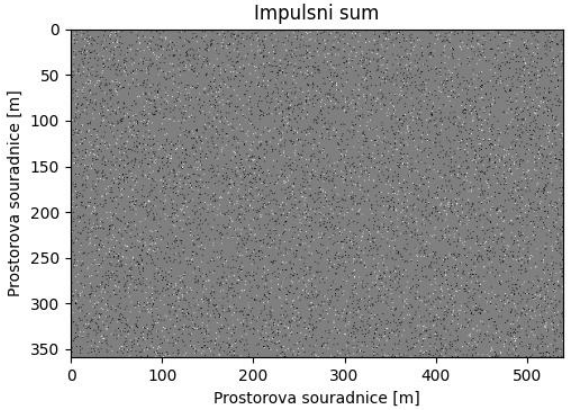


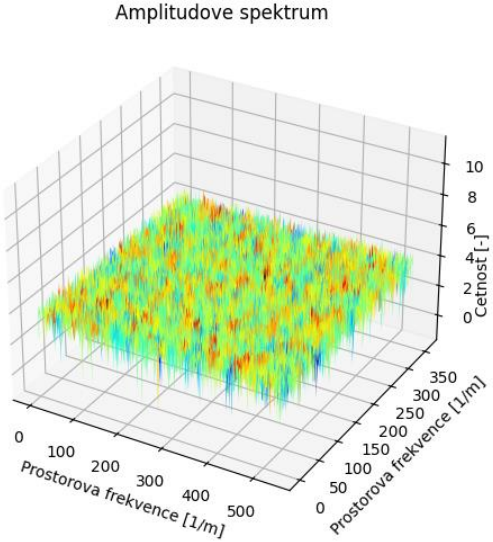


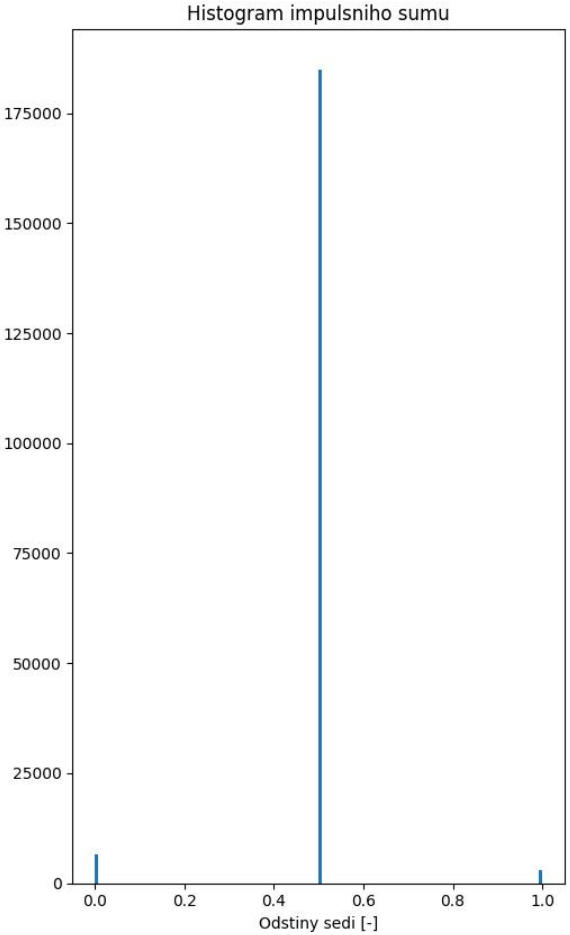
Lokální 2D operátory



Úkol 2

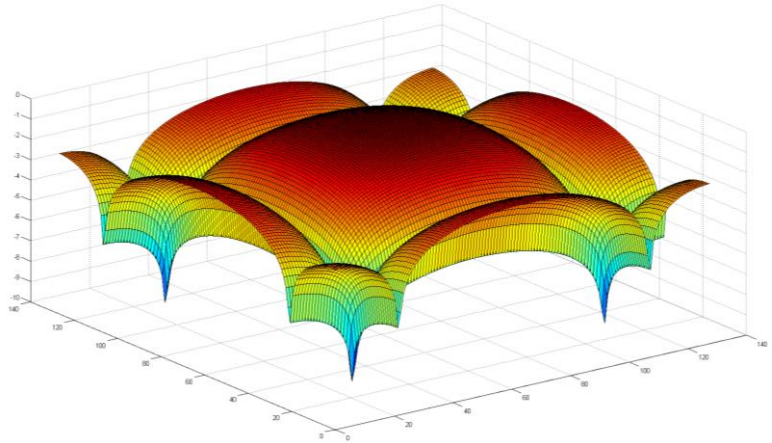

Impulsni sum
Prostorova souradnice [m]
Prostorova souradnice [m]
0
50
100
150
200
250
300
350
0
100
200
300
400
500


Amplitudove spektrum
Prostorova frekvence [1/m]
Prostorova frekvence [1/m]
Cetnost [-]
0
100
200
300
400
500
0
50
100
150
200
250
300
350
0
2
4
6
8
10

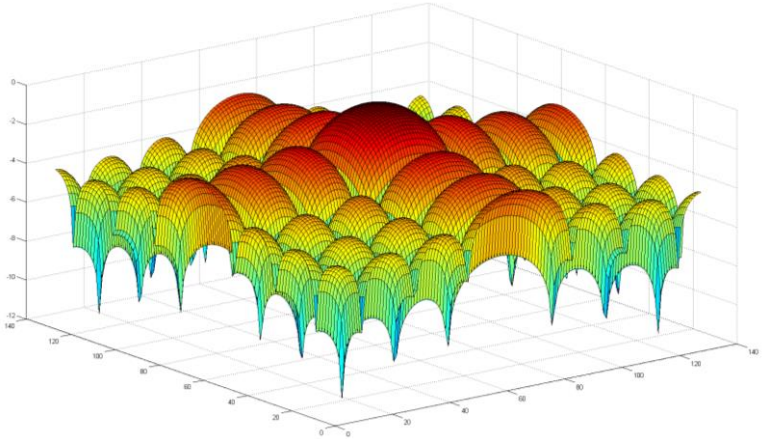

Histogram impulsniho sumu
Odstiny sedi [-]
0
25000
50000
75000
100000
125000
150000
175000
0.0
0.2
0.4
0.6
0.8
1.0

Potlačení šumu :

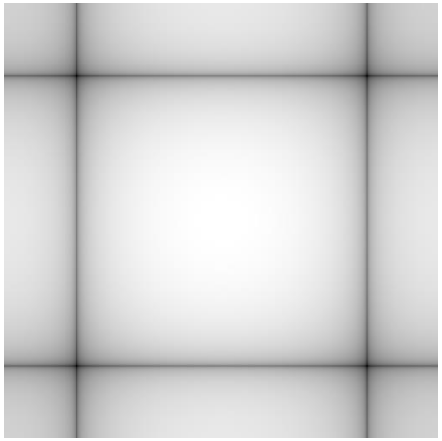
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \cdot \frac{1}{9}$$

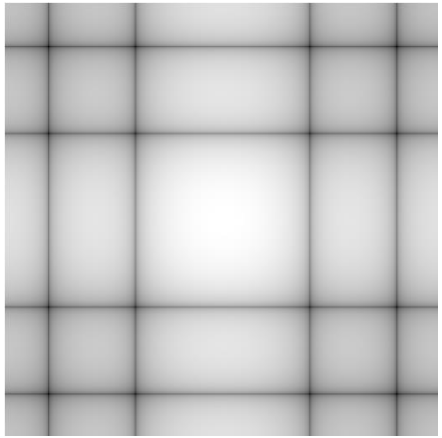


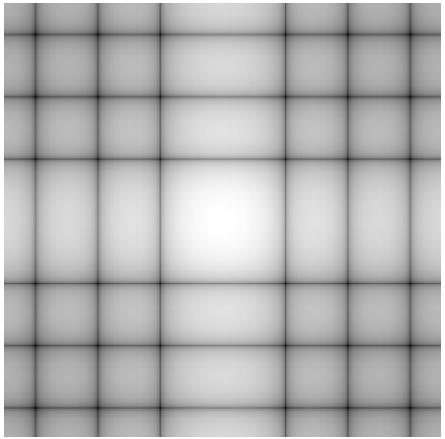
$$\begin{bmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{bmatrix} \cdot \frac{1}{K}$$

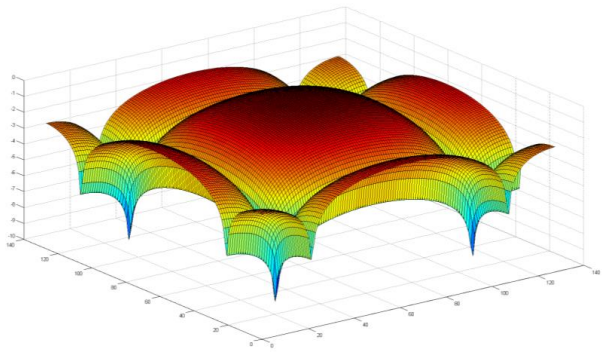


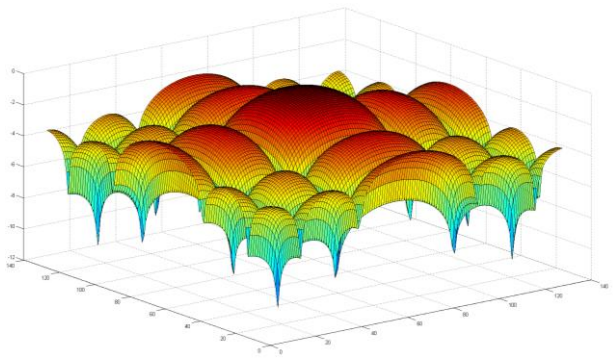
Frekvenční charakteristiky 2D průměrujících filtrů

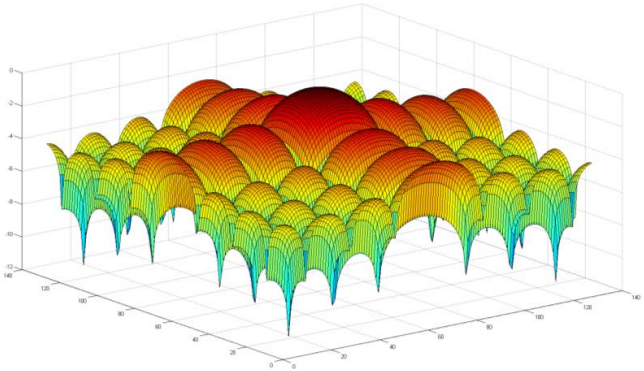




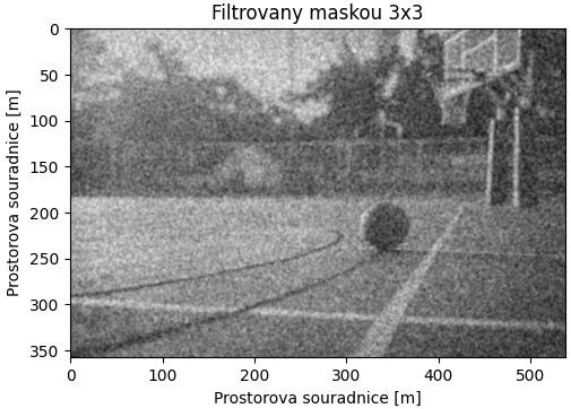


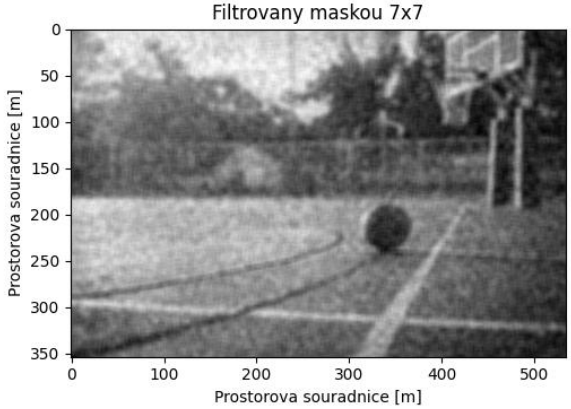


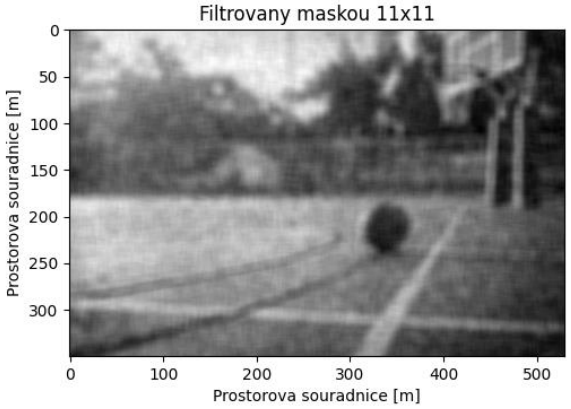


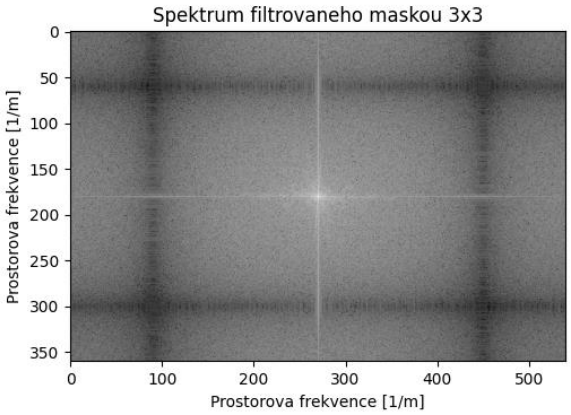


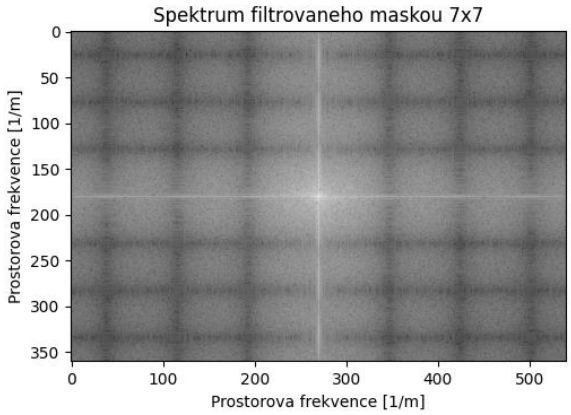
Úkol 3

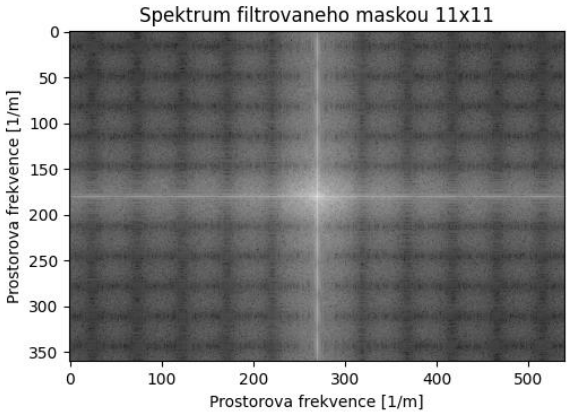

Filtrovany maskou 3x3
Prostorova souradnice [m]
Prostorova souradnice [m]


Filtrovany maskou 7x7
Prostorova souradnice [m]
Prostorova souradnice [m]


Filtrovany maskou 11x11
Prostorova souradnice [m]
Prostorova souradnice [m]


Spektrum filtrovaneho maskou 3x3
Prostorova frekvence [1/m]
Prostorova frekvence [1/m]

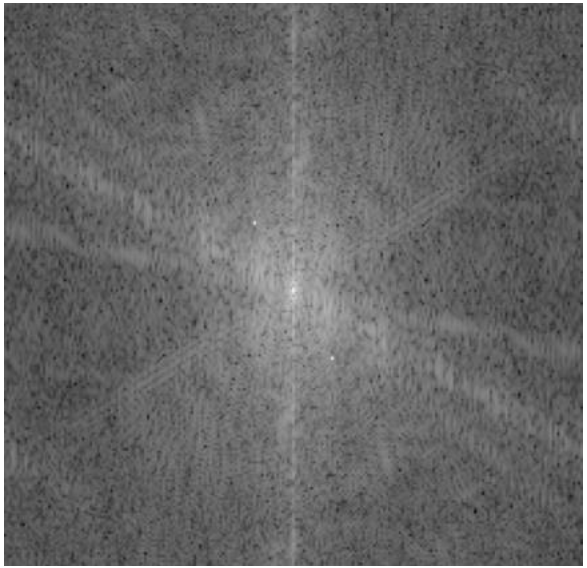

Spektrum filtrovaneho maskou 7x7
Prostorova frekvence [1/m]
Prostorova frekvence [1/m]


Spektrum filtrovaneho maskou 11x11
Prostorova frekvence [1/m]
Prostorova frekvence [1/m]

Šum typu moiré

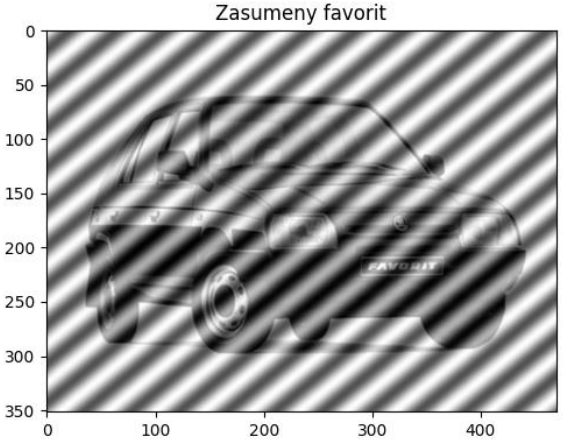


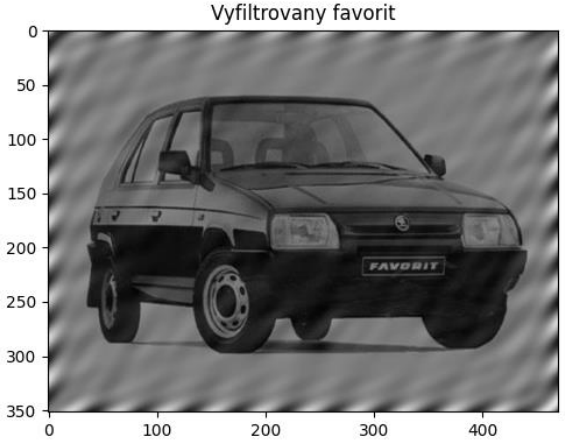


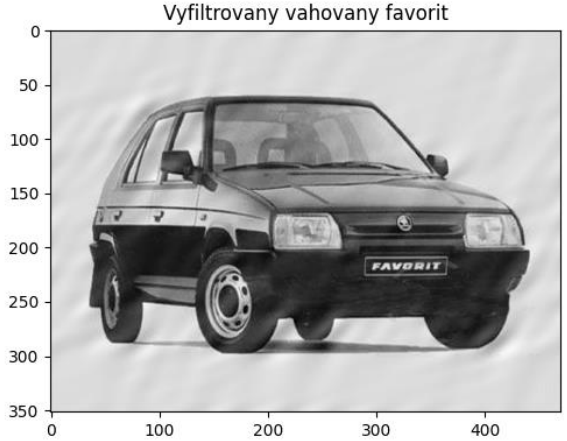


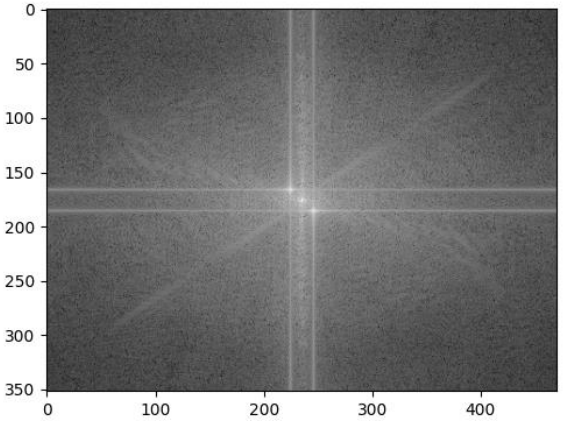
Úkol 4

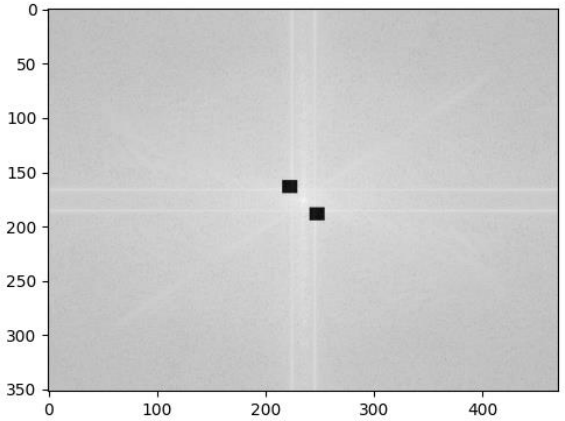
Lokální 2D operátory

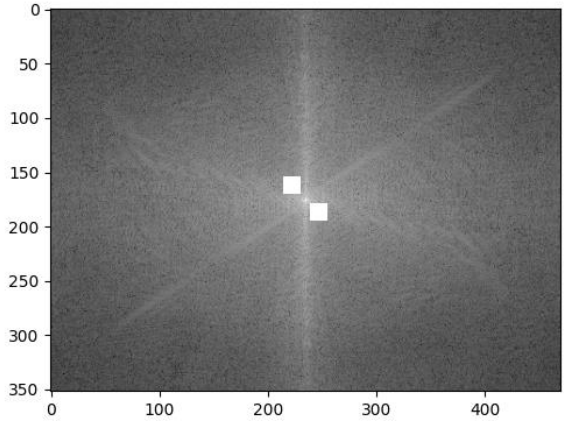












Závěr cvičení

- git add .
- git commit –m „message“
- git push origin main